# Angle bisectors and perpendicular bisectors

**1** Complete the following statements:

**a** A point is on the perpendicular bisector of a line segment if and only if . . .

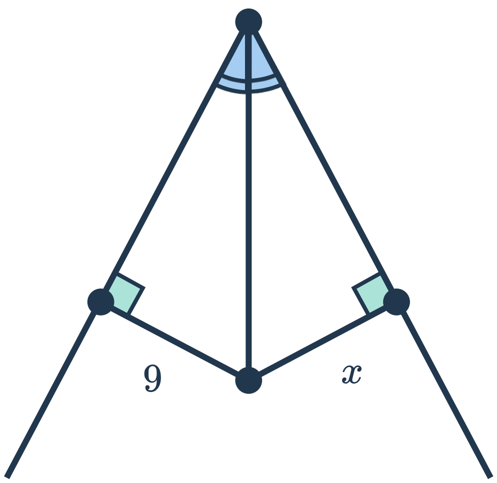
**b** A point is on the bisector of an angle if and only if . . .

**2** For each of the following diagrams:

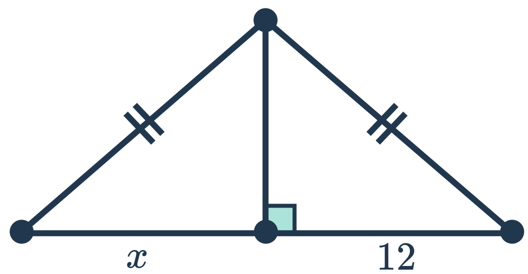
**i** Find $x$.

**ii** Justify your answer.

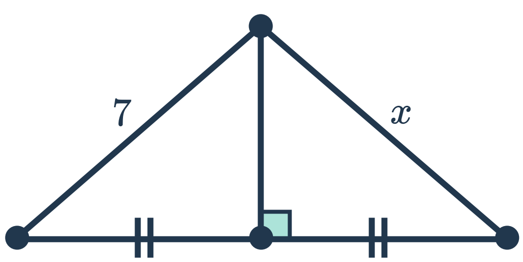
**a**



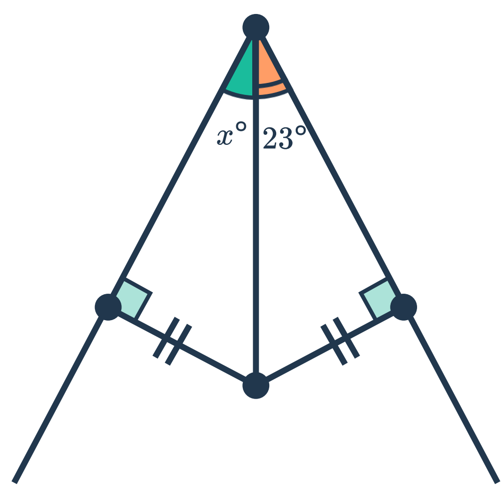

**b**




**c**




**d**



3 For each of the following triangle types, co[redacted]t the perpendicular bisectors of the three sides:

a Acute triangle
b Right triangle
c Obtuse triangle
d Equilateral triangle

4 For each of the following triangle types, construct the angle bisectors of the three angles:

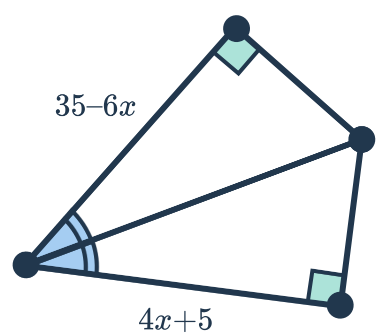
a Acute triangle
b Right triangle
c Obtuse triangle
d Equilateral triangle

5 For each of the following diagrams:

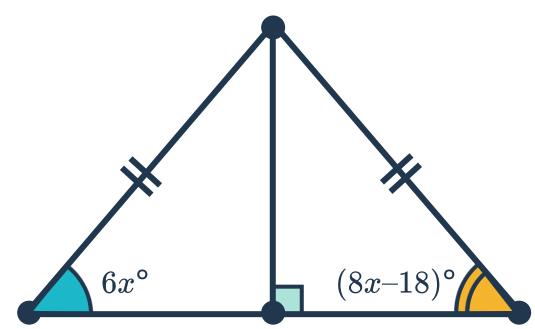
i State the theorem and congruence test used to prove the two triangles are congruent.

ii Find $x$.

a




b

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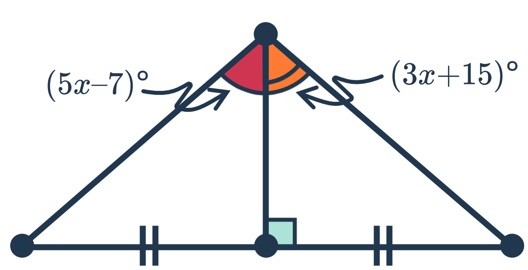


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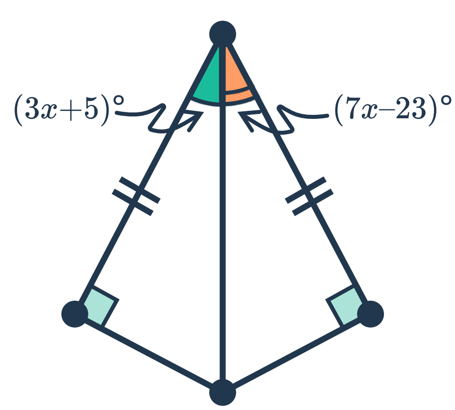
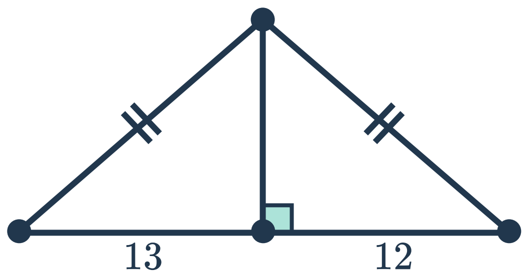
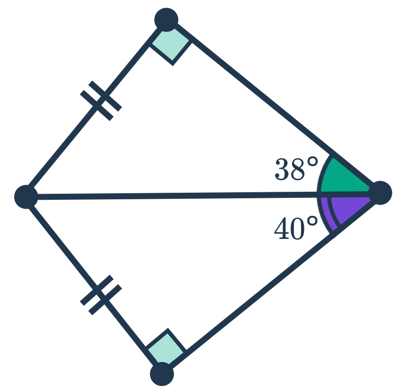


**6** Determine whether each of the following diagrams is valid. If not, identify the contradiction.

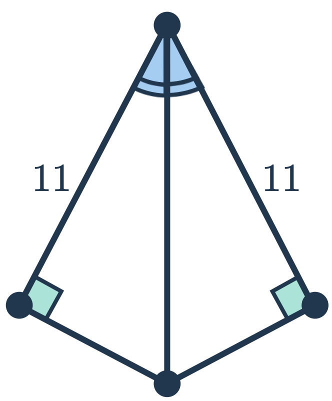
**a**



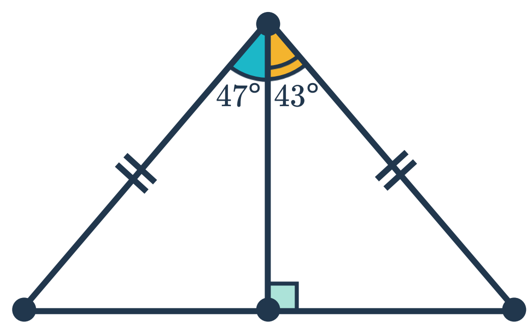

**b**




**c**



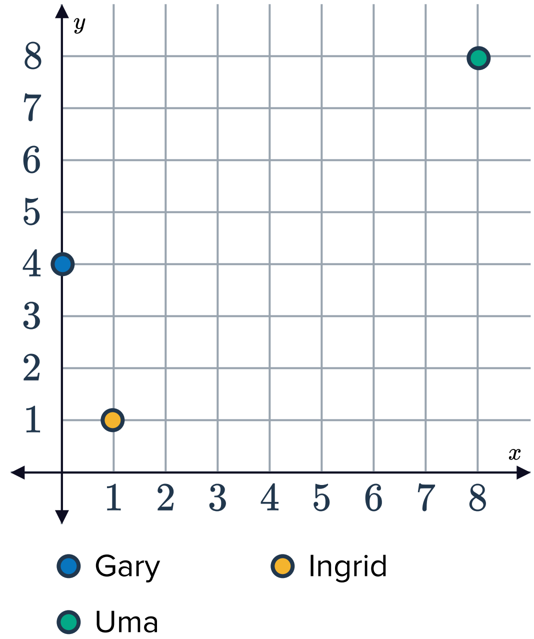

**d**



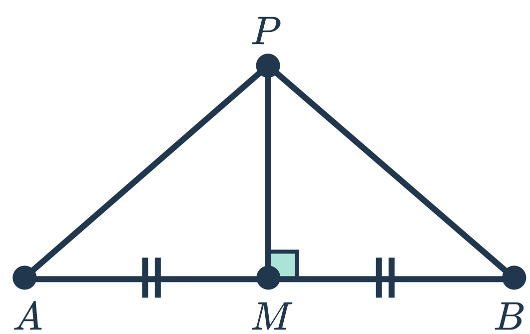
7 Gary, Uma and Ingrid are deciding on a location to meet up at during the holidays, such that the meeting place is equidistant from each of their homes.

a How can they find the exact coordinates for such a meeting place?

b Copy the diagram and make the appropriate constructions to find the meeting place.



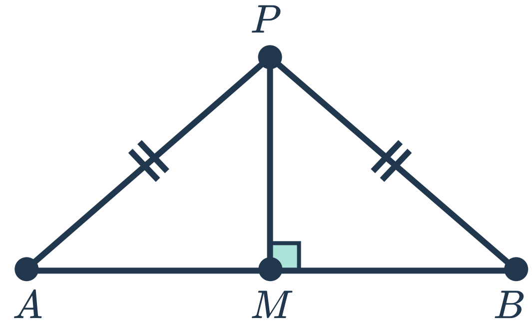

8 Construct a proof of the perpendicular bisector theorem.
GIven: $\overline{PM}$ is the perpendicular bisector of $\overline{AB}$
Prove: $PA = PB$




9 Construct a proof of the converse of perpendicular bisector theorem.
Given:

- $PA = PB$
- $\overline{PM} \perp \overline{AB}$

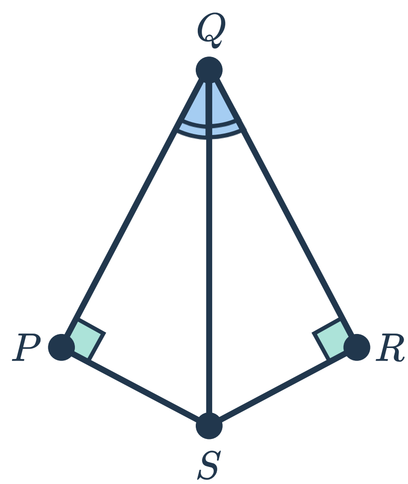
Prove: $AM = BM$



**10** Construct a proof of the angle bisector theorem.
Given:

- $\overline{QS}$ bisects $\angle PQR$
- $\overline{SP} \perp \overline{PQ}$
- $\overline{SR} \perp \overline{RQ}$

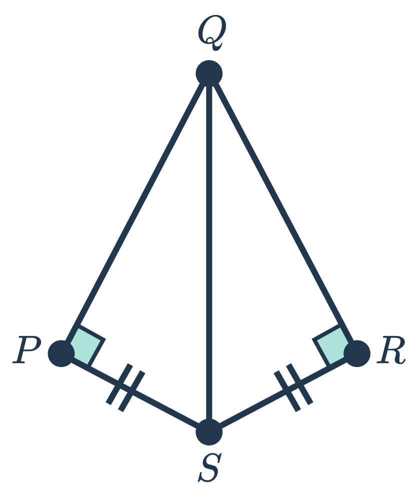
Prove: $SP = SR$



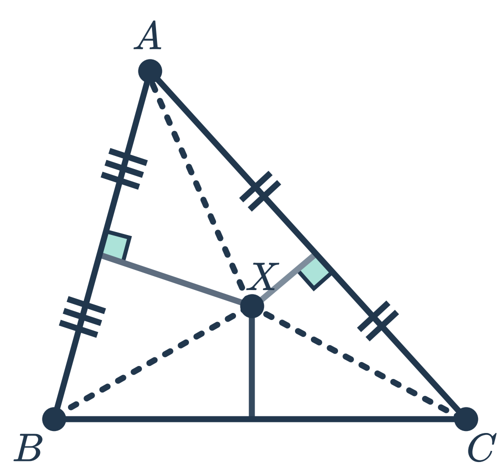

**11** Construct a proof of the converse of angle bisector theorem.
Given:

- $SP = SR$
- $\overline{SP} \perp \overline{PQ}$
- $\overline{SR} \perp \overline{RQ}$

Prove: $m\angle PQS = m\angle RQS$




**12** Given that the perpendicular bisectors of $\overline{AB}$ and $\overline{AC}$ intersect at the point $X$, prove that $X$ lies on the perpendicular bisector of $\overline{BC}$.



**13** Given that the angle bisectors of $\angle ABC$ and $\angle BAC$ intersect at the point $X$, prove that $X$ lies on the angle bisector of $\angle ACB$.